THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 11: **Areas Related to Circles** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 77 cm²	$\frac{1}{8} \times \frac{22}{7} \times 196 = 77$
2	**(a)** $\frac{22}{3}$ cm	$\frac{1}{12} \times 2 \times \frac{22}{7} \times 14$
3	**(a)** $\frac{77}{8}$ cm²	$r = 3.5$; $\frac{1}{4} \times \frac{22}{7} \times 12.25$
4	**(a)** 30°	$\frac{360^\circ}{12}$
5	**(b)** Incorrect — that is the arc length; area = $\frac{\theta}{360^\circ}\pi r^2$	Formula mix-up
6	**(a)** 30°	6° per minute
7	**(a)** Student A is correct	Major sector angle $= 300^\circ$
8	**(b)** 36 cm	$\pi r + 2r = 22 + 14$
9	**(a)** 25π cm²	Area $= \frac{1}{2}lr$
10	**(a)** $\frac{\pi r^2}{4} - \frac{r^2}{2}$	Sector − right triangle

SECTION — B (Very Short Answer) — Answers 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{30}{360} \times 3.14 \times 16 = 4.19$ cm² (approx.)	1+1
12	Angle 30°: $\frac{1}{12} \times \frac{22}{7} \times 196 = \frac{154}{3} \approx 51.33$ cm²	1+1
13	$\frac{1}{8} \times \frac{22}{7} \times 2025 = \frac{22275}{28} \approx 795.54$ cm²	1+1
14	Arc $= \frac{1}{6} \times 2 \times \frac{22}{7} \times 21 = 22$ cm > 21 cm. Claim is **correct**.	1+1

SECTION — C (Short Answer) — Answers 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sector $= 78.5$, triangle $= 50 \Rightarrow$ minor $= 28.5$ cm²; major $= 314 - 28.5 = 285.5$ cm²	1+1+1
16	$\frac{80}{360} \times 3.14 \times 16.5^2 = \frac{2}{9} \times 854.865 \approx 189.97$ km²	1+1+1
17	(i) $\frac{1}{5} \times 220 = 44$ cm (ii) $\frac{1}{5} \times 3850 = 770$ cm² (iii) $44 + 70 = 114$ cm	1+1+1
	OR: Sector $= \frac{1}{3} \times 3.14 \times 144 = 150.72$; triangle $= \frac{1}{2}r^2 \sin 120^\circ = 36\sqrt{3} = 62.28$; segment $= 88.44$ cm²	1+1+1
18	$8.8 = \frac{30}{360} \times 2 \times \frac{22}{7} \times r \Rightarrow r = 16.8$ cm	1+1+1
	OR: Arc $= 16.4 - 10.4 = 6$ cm; area $= \frac{1}{2} \times 6 \times 5.2 = 15.6$ cm²	1+1+1

SECTION — D (Long Answer) — Answers 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{1}{4} \times 176 = 44$ cm (ii) $\frac{1}{4} \times 2464 = 616$ cm² (iii) $616 - 392 = 224$ cm² (iv) $2464 - 224 = 2240$ cm²	1+1+2+1
	OR: (i) 120°: $\frac{1}{3} \times 1386 = 462$ cm² (ii) 60°: $\frac{1}{6} \times 88 = \frac{44}{3} \approx 14.67$ cm (iii) 90°: $\frac{1}{4} \times 616 = 154$ cm²	2+2+1
20	(i) Quadrant: $\frac{1}{4} \times \frac{22}{7} \times 196 = 154$ m² (ii) $\frac{1}{4} \times \frac{22}{7} \times 256 \approx 201.14$; increase ≈ 47.14 m² (iii) $320 - 154 = 166$ m²	2+2+1
	OR: $22 = \frac{\theta}{360} \times 88 \Rightarrow \theta = 90^\circ$ (1). Sector $= 154$, triangle $= 98 \Rightarrow$ minor segment $= 56$ cm² (3); major $= 616 - 56 = 560$ cm² (1)	1+3+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME — SET B

Chapter 11: **Areas Related to Circles** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 462 cm²	$\frac{1}{3} \times 1386$
2	**(a)** 11 cm	$\frac{1}{4} \times 44$
3	**(a)** 38.5 cm²	$r = 7$; $\frac{154}{4}$
4	**(a)** 120°	$20 \times 6^\circ$
5	**(b)** Incorrect — the area becomes 4 times	Area $\propto r^2$
6	**(a)** 22 cm	$\frac{1}{8} \times 176$
7	**(a)** Student A is correct	Area $= \frac{1}{2}lr$
8	**(a)** $2r + \frac{\theta}{360^\circ} \times 2\pi r$	Two radii + arc
9	**(a)** 77 cm²	$\frac{1}{2} \times 154$
10	**(c)** 60°	$\frac{\theta}{360} \times 36\pi = 6\pi$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{1}{6} \times \frac{22}{7} \times 110.25 = 57.75$ cm²	1+1
12	$11 = \frac{\theta}{360} \times 44 \Rightarrow \theta = 90^\circ$	1+1
13	Quadrant: $\frac{1}{4} \times \frac{22}{7} \times 49 = 38.5$ m²	1+1
14	38.5 cm² is the sector. Segment $= 38.5 - \frac{1}{2} \times 7 \times 7 = 38.5 - 24.5 = 14$ cm². Claim is false.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sector $= \frac{308}{3} \approx 102.67$; equilateral triangle $= \frac{\sqrt{3}}{4} \times 196 = 84.77$ ⇒ segment ≈ 17.90 cm²	1+1+1
16	(i) $\frac{1}{9} \times 1386 = 154$ m² (ii) arc $= \frac{1}{9} \times 132 \approx 14.67$; fencing $= 14.67 + 42 = 56.67$ m	1½+1½
17	Sector $= 346.5$, triangle $= 220.5$ ⇒ minor $= 126$ cm²; major $= 1386 - 126 = 1260$ cm²	1+1+1
	OR: $154 = \frac{\theta}{360} \times 616 \Rightarrow \theta = 90^\circ$; arc $= \frac{1}{4} \times 88 = 22$ cm	1½+1½
18	$1\frac{1}{2}$ h ⇒ 45°: area $= \frac{1}{8} \times \frac{22}{7} \times 36 \approx 14.14$ cm²; arc $= \frac{1}{8} \times \frac{264}{7} = \frac{33}{7} \approx 4.71$ cm	2+1
	OR: $\frac{1}{8} \times 2464 = 308$ m²; cost = ₹1232	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{1}{3} \times 88 \approx 29.33$ cm (ii) $\frac{616}{3} \approx 205.33$ cm² (iii) triangle $= \frac{1}{2}r^2 \sin 120^\circ = 49\sqrt{3} = 84.77$ ⇒ 120.56 cm² (iv) $616 - 120.56 = 495.44$ cm²	1+1+2+1
	OR: (i) $\frac{3}{8} \times 2464 = 924$ cm² (ii) $\frac{3}{8} \times 176 = 66$ cm (iii) $132 + 462 = ₹594$	2+1+2
20	(i) 60° (ii) $\frac{1}{6} \times 1386 = 231$ cm² (iii) $\frac{1}{6} \times 132 = 22$ cm (iv) $2 \times 231 = 462$ ✓	1+2+1+1
	OR: (i) $\frac{5}{12} \times 132 = 55$ cm (ii) $\frac{5}{12} \times 1386 = 577.5$ cm² (iii) $1386 - 577.5 = 808.5$ cm²	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.